Module 7: Knowledge Graphs for Biomedical Data Integration

Disease: Ovarian Cancer

1. Knowledge Graph Design and Evidence Integration

A biomedical knowledge graph was built to combine ovarian cancer multi-omics results with biological pathways, clinical symptoms, drug-target relationships, and genetic variants. The final graph contained 24 nodes and 29 relationships, including eight genes, five phenotypes, four pathways, four drugs, two diseases, and one variant. Seven ovarian cancer genes (TP53, RB1, EGFR, ERBB2, MKI67, PTEN, and PIK3CA) were selected from the Module 2 analysis. ESR1 was added separately as a breast cancer-related gene for the cross-disease query.

Six node types were connected through five relationships: ASSOCIATED_WITH, HAS_PHENOTYPE, PARTICIPATES_IN, TARGETS, and OCCURS_IN. This structure connects molecular evidence with biological functions and potential drug targets. Gene symbols were kept consistent to avoid duplicate nodes, and all 29 relationships connected existing nodes. The original Module 2 integrated scores were kept as gene properties. Missing GWAS p-values and RNA or protein log fold changes were left blank rather than estimated from other scores.

Each relationship included information about its source and confidence level. Gene-disease associations came from the Module 2 multi-omics analysis. Pathway, phenotype, and drug-target relationships were added based on known biology, but some still require further verification. A drug targeting a particular gene was not considered evidence that the drug is effective against ovarian cancer.

2. Biomarker Prioritization and Biological Interpretation

PageRank was used to measure how strongly each gene was connected within the network. The PageRank values were then combined with the Module 2 integrated scores using equal weights:

Combined score = 0.5 × normalized PageRank + 0.5 × integrated evidence score

Table 1. Ovarian Cancer candidate-biomarker ranking

Rank	Gene	PageRank	Integrated score	Combined score
1	TP53	0.0413	0.9913	0.9957
2	EGFR	0.0413	0.9452	0.9726
3	ERBB2	0.0413	0.9014	0.9507
4	PIK3CA	0.0413	0.6370	0.8185
5	RB1	0.0223	0.9796	0.4898

6	MKI67	0.0223	0.8595	0.4298
7	PTEN	0.0223	0.8248	0.4124

TP53 ranked first (0.9957), which is consistent with its role in tumor suppression, DNA damage response, and cell-cycle regulation. EGFR and ERBB2 followed, partly because of their connections to receptor signaling pathways and drugs. PIK3CA ranked fourth despite having a lower original integrated score, showing how network connections can affect the final ranking. RB1 showed an interesting difference between the two scoring methods. Although it had the second-highest integrated score (0.9796), it dropped to fifth place after PageRank was included. This was likely because RB1 had fewer connections in the graph, not because it was less important in ovarian cancer. Also, normalized PageRank divided the genes into only two groups (0 and 1). This made the ranking highly dependent on the graph structure and the chosen 50:50 weighting. Therefore, the combined scores should be used to compare candidates within this analysis rather than to predict their clinical performance.

3. Drug Queries and Path-Based Hypothesis Generation

The two-hop query identified three drug-target relationships: Alpelisib–PIK3CA, Gefitinib–EGFR, and Trastuzumab–ERBB2. These results show how the graph connects ovarian cancer-related genes to existing drugs. However, these drugs were listed as approved for other conditions, so their presence in the graph does not mean they are effective treatments for ovarian cancer. The multi-hop query identified the following path between two diseases:

Fulvestrant → ESR1 → Breast cancer → Fatigue ← Ovarian cancer

This path connects a breast cancer drug to ovarian cancer through a shared symptom. However, fatigue is common in many diseases and does not provide strong biological evidence for drug repurposing. A more specific hypothesis is that some ovarian tumors with active estrogen receptor signaling might respond to endocrine therapy. This could be tested by measuring ESR1 expression and pathway activity in ovarian tumor samples, then comparing responses to fulvestrant in estrogen receptor-positive and receptor-negative ovarian cancer models. Cell viability, proliferation, and changes in gene expression could be measured to evaluate the response.

4. Community Detection and Network Structure

Community detection identified five groups within the graph. One group contained TP53, RB1, MKI67, rs28934578, and two cell-cycle-related pathways. This grouping makes biological sense because TP53 and RB1 regulate cell-cycle progression, while MKI67 is commonly used as a

marker of cell proliferation. The TP53 variant rs28934578 was also connected to this group, although its specific relevance to ovarian cancer was not established in this analysis.
Two other groups were associated with signaling pathways and drug targets. The ERBB receptor signaling group included EGFR, ERBB2, Gefitinib, and Trastuzumab. The PI3K/AKT group contained PIK3CA, PTEN, and Alpelisib. The remaining groups connected ovarian cancer to its clinical symptoms and breast cancer to ESR1, Fulvestrant, and fatigue. These groups reflect the relationships included in the graph and do not necessarily represent distinct molecular subtypes of ovarian cancer.

5. Limitations and Future Directions

One major limitation was that not all relationships had the same level of supporting evidence. The gene-disease associations came from the Module 2 multi-omics analysis, while some phenotype, pathway, and drug-target relationships still need to be verified using databases such as Reactome, Open Targets, ChEMBL, ClinVar, and the Human Phenotype Ontology. The connection between ovarian cancer and breast cancer was particularly weak because it depended on fatigue rather than a shared molecular pathway. More specific biological evidence could lead to different drug repurposing results.
The graph included only seven ovarian cancer genes and one variant, which limited the number of possible connections and may have affected PageRank and community detection. The variant was not obtained from an ovarian cancer patient dataset. Patient stratification was not performed because patient-level data were unavailable.
AI-assisted coding was used to modify the starter notebook, organize the graph, and combine PageRank with the Module 2 scores. The results were checked against the original CSV, node and relationship counts, query outputs, and community assignments. Future work should include verified Ensembl IDs, stronger evidence for individual relationships, more ovarian cancer variants, and comparisons using different scoring weights.
Overall, the knowledge graph brought together molecular, clinical, and drug-related information in a single network. TP53 ranked highest, while the ERBB and PI3K/AKT groups showed possible targets for further investigation. The analysis also showed that finding a connection in a graph is only the first step. Whether that connection is biologically meaningful depends on the

evidence supporting it.

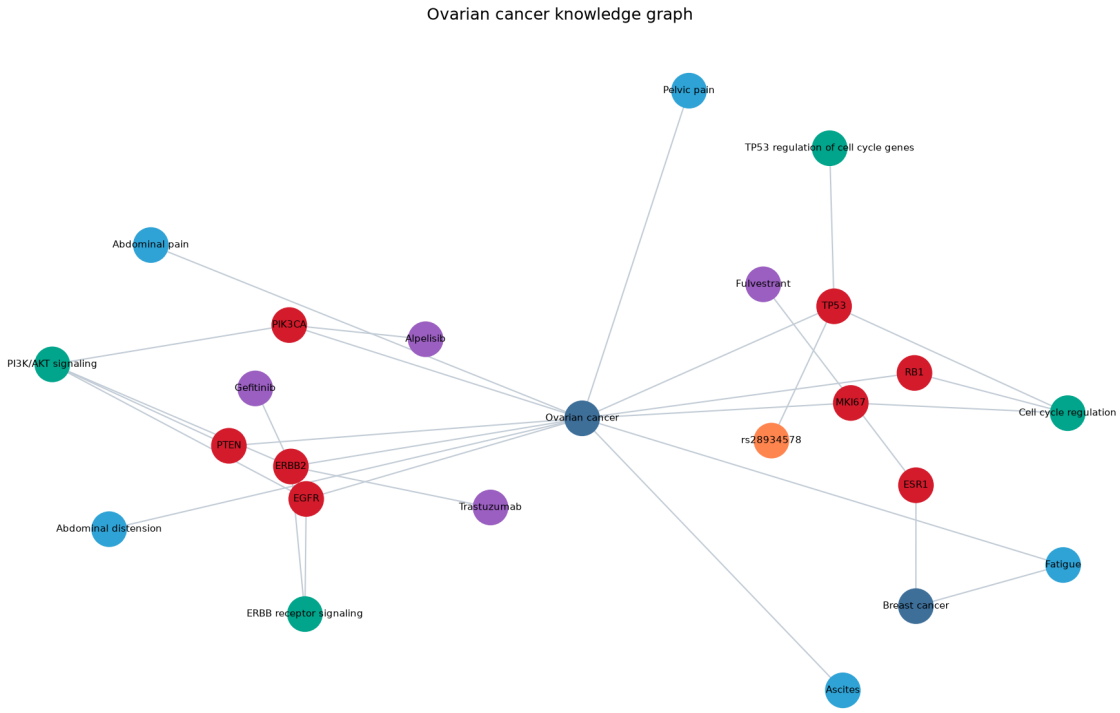


Figure 1. Ovarian cancer knowledge graph integrating disease, gene, phenotype, pathway, drug, and variant nodes. Node colors indicate entity types, and edges represent curated biological or pharmacological relationships.